ecbench report: ECBS1h88c58e3e8c89

Label: comprehensive generic sweep, Koblitz K_1 over F_2^n, n in {17,19,23,29,43,47}

Description: Method × size scaling study (koblitz curves, degrees n of K_a/F_2^n 17,19,23,29,43,47; 2 independent single-target workload(s) per curve, 3 interleaved round(s)). One curve per size so `ecbench bound fit` and `ecbench report` can fit cost against r. Counted group operations; a measurement, not a performance claim.

Generated by `ecbench report` (ecbench.report/v1). Observations only; derived numbers carry the key of their derivation (last section).

Session

field	value
session id	ECBS1h88c58e3e8c89
spec id	ECS1ha3fdd42bc1ef
commit	f793f1fe5edc540668ed836d5e975eb040a34dff (dirty)
status	complete
run options	cpus="auto", lock=/tmp/crypto-bench.lock, wait_for_lock=false, allow_busy=true

Host and isolation

Intel(R) Xeon(R) Processor @ 2.10GHz; 4 logical / 4 physical cores; kernel 6.18.44-fc-v139; 15.7 GiB; env class ECBENV2he1389885ccd0; virtualization cpu_flag=hypervisor, systemd_detect_virt_vm=kvm.

Isolation earned by the 180 measured records: L0 0, L1 180, L2 0, L3 0. Required of a wall-clock figure: L2.

blocker	measured records
L2: preflight not quiet: PSI cpu some avg10 7.02 > 5.00	180
L3: governor is None, not performance	180
L3: not known to be bare metal	180
L3: run CPU is in neither isolcpus nor nohz_full	180
L3: turbo/boost is not known to be off	180
L2: 1 ticks of hypervisor steal on the run CPU	15
L2: 52.3 preemptions per second	1
L2: 53.0 preemptions per second	1
L2: 54.9 preemptions per second	1
L2: 56.0 preemptions per second	1
L2: 61.9 preemptions per second	1
L2: 62.9 preemptions per second	1

Arms

arm	role	method	params	expected S [d4]
rho-signed_frobenius_strong	baseline	rho.signed_frobenius_strong	dp_bits=4; lanes=32; step_cap_factor=2000	0.2149
rho-negation	reference	rho.negation	cap_multiple=64	0.8862
bsgs-negation	candidate	bsgs.negation	defaults	1.0000
kangaroo-vow	candidate	kangaroo.vow	cap_multiple=64	2.0000
rho-negation-aa	control	rho.negation	cap_multiple=64	0.8862

Workloads

workload	curve	family	log2 r	A	target index	target seed
W10f4a21252f1	icv1-f2m17-tm101-00378d4e	koblitz	16.001	34	0	20261002
W782cd1b187b6	icv1-f2m17-tm101-00378d4e	koblitz	16.001	34	1	20261002
Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	koblitz	18.002	38	0	20261002
W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	koblitz	18.002	38	1	20261002
W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	koblitz	22.001	46	0	20261002
W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	koblitz	22.001	46	1	20261002

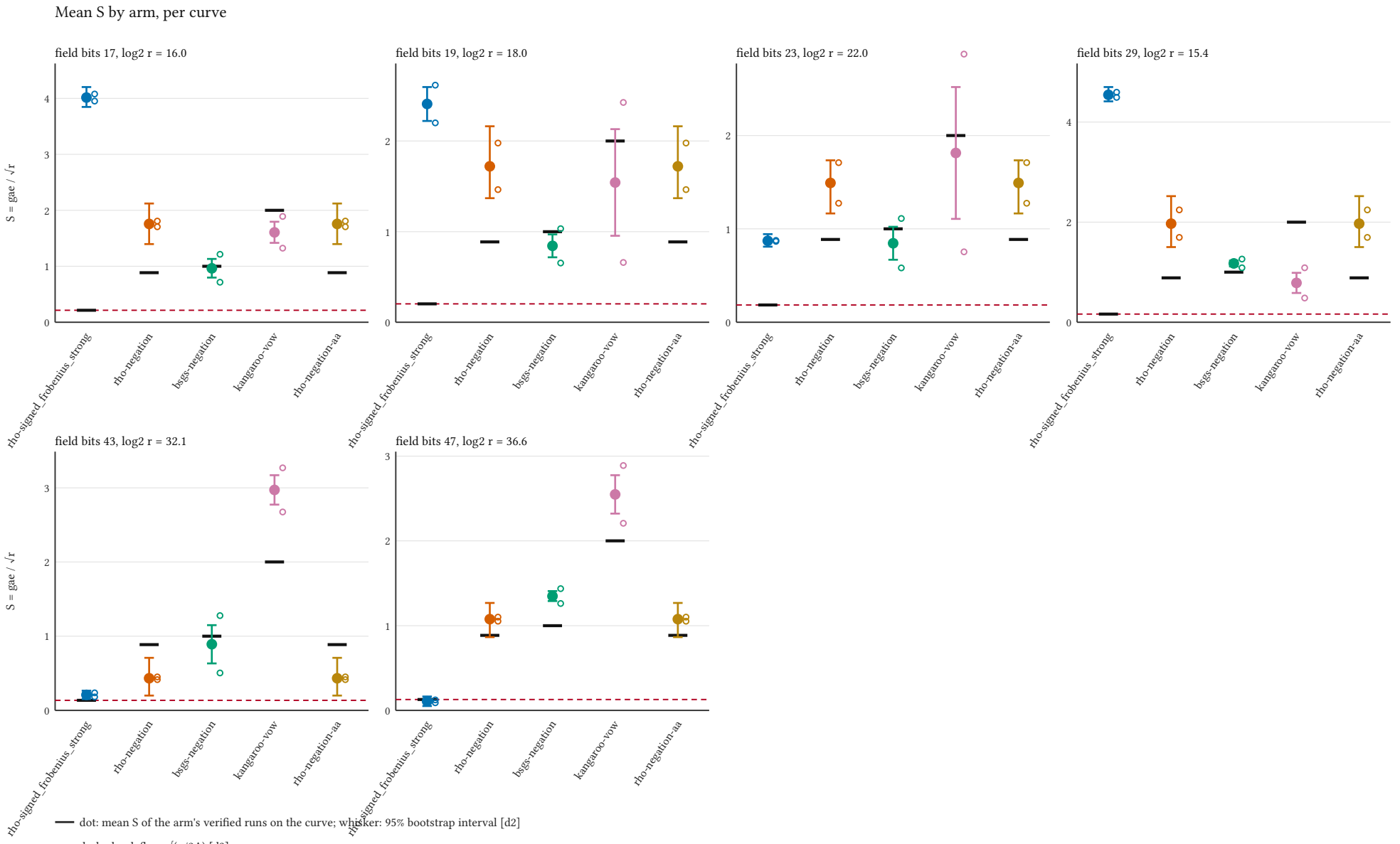
workload	curve	family	log2 r	A	target index	target seed
W4aedbc100172	icv1-f2m29-tm40309-30c52b96	koblitz	15.374	58	0	20261002
W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	koblitz	15.374	58	1	20261002
W14592200e40d	icv1-f2m43-tm998717-e2e742b0	koblitz	32.113	86	0	20261002
W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	koblitz	32.113	86	1	20261002
W46a98355e416	icv1-f2m47-t22705043-f4e44623	koblitz	36.636	94	0	20261002
W606d09fccb11	icv1-f2m47-t22705043-f4e44623	koblitz	36.636	94	1	20261002

Runs

Planned 240 (60 warm-up, 180 measured); records 240 (180 measured); verified 180 (180 with matches_target); not verified 0; planned without a record 0.

No record has a status other than verified.

Graphs



Session ECBS1h88c58e3e8c89, n = 180 verified runs. Unit: S = gae / √r.

Figure 1: Mean S by arm per curve with 95% bootstrap whiskers, the floor and expected S; session ECBS1h88c58e3e8c89, n = 180 verified runs, unit gae (S = gae / √r)

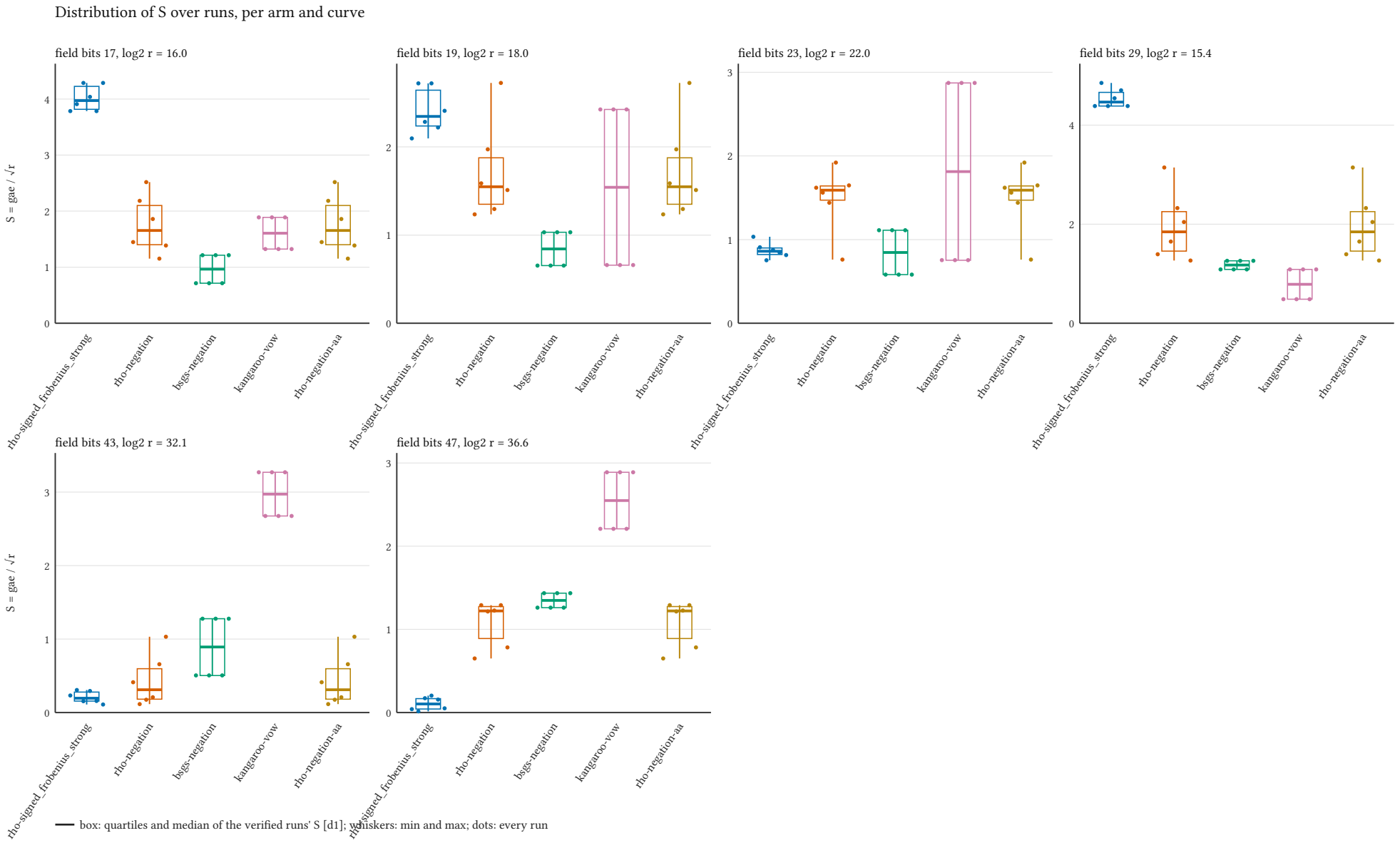
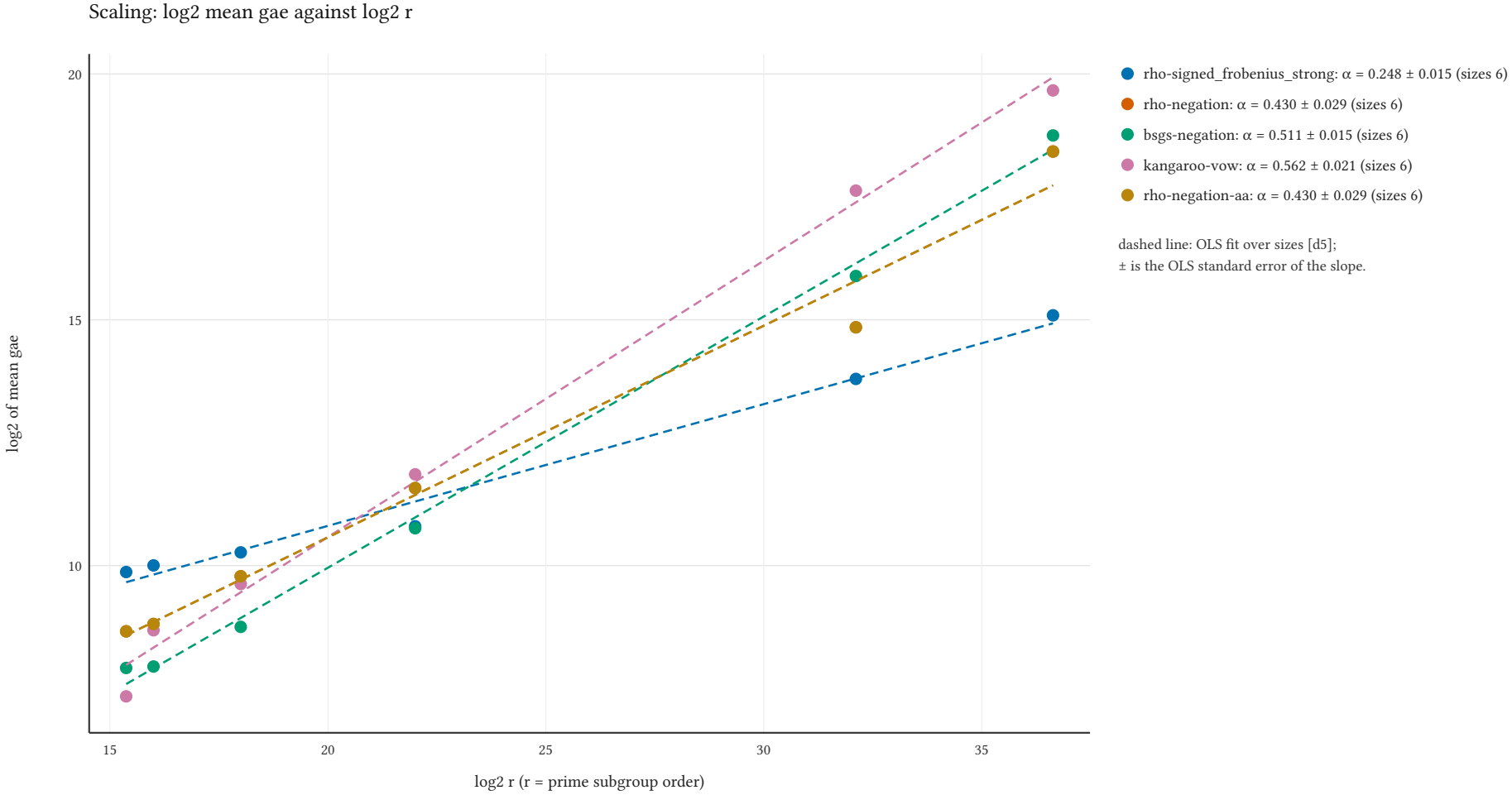


Figure 2: Distribution of S over the verified runs, per arm and curve; session ECBS1h88c58e3e8c89, n = 180 verified runs, unit gae (S = gae / √r)



Session ECBS1h88c58e3e8c89, n = 180 verified runs over 6 sizes. Unit: gae (counted group operations).

Figure 3: log2 mean gae against log2 r with OLS fits and fitted exponents; session ECBS1h88c58e3e8c89, n = 180 verified runs, unit gae (S = gae / √r)

Results per arm and curve

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-signed_frobenius_strong	icv1-f2m17-tm101-00378d4e	16.001	6 / 6	4.0149	3.9733	3.7839	4.2876	0.2312	[3.8470, 4.2037]	0.2149	0.2149
rho-signed_frobenius_strong	icv1-f2m19-tm797-9c54981b	18.002	6 / 6	2.4089	2.3471	2.0963	2.7208	0.2620	[2.2215, 2.5962]	0.2033	0.2033
rho-signed_frobenius_strong	icv1-f2m23-tm5197-1f85e9e1	22.001	6 / 6	0.8716	0.8611	0.7522	1.0329	0.0955	[0.8092, 0.9444]	0.1848	0.1848
rho-signed_frobenius_strong	icv1-f2m29-tm40309-30c52b96	15.374	6 / 6	4.5440	4.4655	4.3878	4.8538	0.1972	[4.4137, 4.6993]	0.1646	0.1646
rho-signed_frobenius_strong	icv1-f2m43-tm998717-e2e742b0	32.113	6 / 6	0.2089	0.1948	0.1088	0.3061	0.0810	[0.1523, 0.2670]	0.1351	0.1351

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-signed_frobenius_strong	icv1-f2m47-t22705043-f4e44623	36.636	6 / 6	0.1070	0.1040	0.0152	0.2044	0.0799	[0.0509, 0.1640]	0.1293	0.1293
rho-negation	icv1-f2m17-tm101-00378d4e	16.001	6 / 6	1.7578	1.6537	1.1519	2.5186	0.5235	[1.3972, 2.1229]	0.2149	0.8862
rho-negation	icv1-f2m19-tm797-9c54981b	18.002	6 / 6	1.7213	1.5486	1.2354	2.7264	0.5574	[1.3700, 2.1624]	0.2033	0.8862
rho-negation	icv1-f2m23-tm5197-1f85e9e1	22.001	6 / 6	1.4919	1.5911	0.7610	1.9203	0.3914	[1.1652, 1.7351]	0.1848	0.8862
rho-negation	icv1-f2m29-tm40309-30c52b96	15.374	6 / 6	1.9704	1.8466	1.2667	3.1449	0.6988	[1.5004, 2.5188]	0.1646	0.8862
rho-negation	icv1-f2m43-tm998717-e2e742b0	32.113	6 / 6	0.4332	0.3104	0.1144	1.0317	0.3547	[0.1996, 0.7080]	0.1351	0.8862
rho-negation	icv1-f2m47-t22705043-f4e44623	36.636	6 / 6	1.0762	1.2215	0.6503	1.2912	0.2835	[0.8639, 1.2672]	0.1293	0.8862
bsgs-negation	icv1-f2m17-tm101-00378d4e	16.001	6 / 6	0.9645	0.9645	0.7146	1.2144	0.2738	[0.7979, 1.1311]	0.2149	1.0000
bsgs-negation	icv1-f2m19-tm797-9c54981b	18.002	6 / 6	0.8431	0.8431	0.6538	1.0324	0.2074	[0.7169, 0.9693]	0.2033	1.0000
bsgs-negation	icv1-f2m23-tm5197-1f85e9e1	22.001	6 / 6	0.8464	0.8464	0.5814	1.1115	0.2904	[0.6697, 1.0231]	0.1848	1.0000
bsgs-negation	icv1-f2m29-tm40309-30c52b96	15.374	6 / 6	1.1745	1.1745	1.0871	1.2618	0.0957	[1.1162, 1.2327]	0.1646	1.0000
bsgs-negation	icv1-f2m43-tm998717-e2e742b0	32.113	6 / 6	0.8913	0.8913	0.5051	1.2776	0.4231	[0.6338, 1.1488]	0.1351	1.0000
bsgs-negation	icv1-f2m47-t22705043-f4e44623	36.636	6 / 6	1.3489	1.3489	1.2618	1.4361	0.0954	[1.2909, 1.4070]	0.1293	1.0000
kangaroo-vow	icv1-f2m17-tm101-00378d4e	16.001	6 / 6	1.6068	1.6068	1.3237	1.8899	0.3101	[1.4181, 1.7955]	0.2149	2.0000
kangaroo-vow	icv1-f2m19-tm797-9c54981b	18.002	6 / 6	1.5428	1.5428	0.6597	2.4259	0.9674	[0.9540, 2.1315]	0.2033	2.0000
kangaroo-vow	icv1-f2m23-tm5197-1f85e9e1	22.001	6 / 6	1.8132	1.8132	0.7537	2.8726	1.1606	[1.1068, 2.5195]	0.1848	2.0000
kangaroo-vow	icv1-f2m29-tm40309-30c52b96	15.374	6 / 6	0.7862	0.7862	0.4853	1.0871	0.3296	[0.5856, 0.9868]	0.1646	2.0000
kangaroo-vow	icv1-f2m43-tm998717-e2e742b0	32.113	6 / 6	2.9719	2.9719	2.6742	3.2695	0.3261	[2.7734, 3.1703]	0.1351	2.0000
kangaroo-vow	icv1-f2m47-t22705043-f4e44623	36.636	6 / 6	2.5490	2.5490	2.2088	2.8892	0.3727	[2.3222, 2.7758]	0.1293	2.0000
rho-negation-aa	icv1-f2m17-tm101-00378d4e	16.001	6 / 6	1.7578	1.6537	1.1519	2.5186	0.5235	[1.3972, 2.1229]	0.2149	0.8862
rho-negation-aa	icv1-f2m19-tm797-9c54981b	18.002	6 / 6	1.7213	1.5486	1.2354	2.7264	0.5574	[1.3700, 2.1624]	0.2033	0.8862
rho-negation-aa	icv1-f2m23-tm5197-1f85e9e1	22.001	6 / 6	1.4919	1.5911	0.7610	1.9203	0.3914	[1.1652, 1.7351]	0.1848	0.8862
rho-negation-aa	icv1-f2m29-tm40309-30c52b96	15.374	6 / 6	1.9704	1.8466	1.2667	3.1449	0.6988	[1.5004, 2.5188]	0.1646	0.8862
rho-negation-aa	icv1-f2m43-tm998717-e2e742b0	32.113	6 / 6	0.4332	0.3104	0.1144	1.0317	0.3547	[0.1996, 0.7080]	0.1351	0.8862
rho-negation-aa	icv1-f2m47-t22705043-f4e44623	36.636	6 / 6	1.0762	1.2215	0.6503	1.2912	0.2835	[0.8639, 1.2672]	0.1293	0.8862

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-signed_frobenius_strong	icv1-f2m17-tm101-00378d4e	6 / 6	1028.2	18.6792	–	–	–	–	–	4.8228	31.7975	informative, L1
rho-signed_frobenius_strong	icv1-f2m19-tm797-9c54981b	6 / 6	1234.3	11.8480	–	–	–	–	–	5.3588	34.1420	informative, L1
rho-signed_frobenius_strong	icv1-f2m23-tm5197-1f85e9e1	6 / 6	1785.5	4.7166	–	–	–	–	–	5.2344	33.7715	informative, L1
rho-signed_frobenius_strong	icv1-f2m29-tm40309-30c52b96	6 / 6	936.3	27.6114	–	–	–	–	–	5.1164	35.3270	informative, L1
rho-signed_frobenius_strong	icv1-f2m43-tm998717-e2e742b0	6 / 6	14234.3	1.5455	–	–	–	–	–	6.4172	37.4550	informative, L1
rho-signed_frobenius_strong	icv1-f2m47-t22705043-f4e44623	6 / 6	34951.4	0.8274	–	–	–	–	–	8.8026	40.6370	informative, L1
rho-negation	icv1-f2m17-tm101-00378d4e	6 / 6	450.2	8.1779	17.2	259.0	–	–	–	4.9102	31.8735	informative, L1
rho-negation	icv1-f2m19-tm797-9c54981b	6 / 6	882.0	8.4664	35.5	643.3	–	–	–	5.3866	35.8955	informative, L1
rho-negation	icv1-f2m23-tm5197-1f85e9e1	6 / 6	3056.3	8.0734	77.8	2523.8	–	–	–	6.2589	40.0155	informative, L1
rho-negation	icv1-f2m29-tm40309-30c52b96	6 / 6	406.0	11.9731	13.3	221.0	–	–	–	5.3264	37.9350	informative, L1
rho-negation	icv1-f2m43-tm998717-e2e742b0	6 / 6	29524.7	3.2057	111.3	28024.8	–	–	–	11.7546	46.5900	informative, L1
rho-negation	icv1-f2m47-t22705043-f4e44623	6 / 6	351670.5	8.3252	658.8	338973.3	–	–	–	101.0653	130.0730	informative, L1
bsgs-negation	icv1-f2m17-tm101-00378d4e	6 / 6	247.0	4.4871	–	238.0	–	–	–	5.3934	39.6570	informative, L1
bsgs-negation	icv1-f2m19-tm797-9c54981b	6 / 6	432.0	4.1468	–	422.0	–	–	–	5.2371	36.2535	informative, L1
bsgs-negation	icv1-f2m23-tm5197-1f85e9e1	6 / 6	1734.0	4.5804	–	1722.0	–	–	–	5.1328	34.6130	informative, L1
bsgs-negation	icv1-f2m29-tm40309-30c52b96	6 / 6	242.0	7.1367	–	233.0	–	–	–	5.9282	44.0475	informative, L1
bsgs-negation	icv1-f2m43-tm998717-e2e742b0	6 / 6	60742.5	6.5952	–	60720.5	–	–	–	22.2053	57.3440	informative, L1
bsgs-negation	icv1-f2m47-t22705043-f4e44623	6 / 6	440800.5	10.4352	–	440771.5	–	–	–	159.7264	189.9410	informative, L1
kangaroo-vow	icv1-f2m17-tm101-00378d4e	6 / 6	411.5	7.4755	57.5	383.5	–	–	–	5.7752	35.8705	informative, L1

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
kangaroo-vow	icv1-f2m19-tm797-9c54981b	6 / 6	790.5	7.5881	107.0	757.5	–	–	–	5.4577	35.3045	informative, L1
kangaroo-vow	icv1-f2m23-tm5197-1f85e9e1	6 / 6	3714.5	9.8119	231.0	3675.5	–	–	–	5.5605	34.1330	informative, L1
kangaroo-vow	icv1-f2m29-tm40309-30c52b96	6 / 6	162.0	4.7774	15.0	131.0	–	–	–	0.0680	38.3075	informative, L1
kangaroo-vow	icv1-f2m43-tm998717-e2e742b0	6 / 6	202527.0	21.9896	1578.0	202464.0	–	–	–	43.4078	75.5000	informative, L1
kangaroo-vow	icv1-f2m47-t22705043-f4e44623	6 / 6	832951.0	19.7186	3199.0	832882.0	–	–	–	197.7384	239.1090	informative, L1
rho-negation-aa	icv1-f2m17-tm101-00378d4e	6 / 6	450.2	8.1779	17.2	259.0	–	–	–	4.8156	33.5865	informative, L1
rho-negation-aa	icv1-f2m19-tm797-9c54981b	6 / 6	882.0	8.4664	35.5	643.3	–	–	–	5.7939	35.6120	informative, L1
rho-negation-aa	icv1-f2m23-tm5197-1f85e9e1	6 / 6	3056.3	8.0734	77.8	2523.8	–	–	–	6.4485	35.1500	informative, L1
rho-negation-aa	icv1-f2m29-tm40309-30c52b96	6 / 6	406.0	11.9731	13.3	221.0	–	–	–	5.2310	35.1640	informative, L1
rho-negation-aa	icv1-f2m43-tm998717-e2e742b0	6 / 6	29524.7	3.2057	111.3	28024.8	–	–	–	11.2771	43.8815	informative, L1
rho-negation-aa	icv1-f2m47-t22705043-f4e44623	6 / 6	351670.5	8.3252	658.8	338973.3	–	–	–	94.8189	127.8885	informative, L1

Results per arm and workload

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-signed_frobenius_strong	W10f4a21252f1	icv1-f2m17-tm101-00378d4e	3 / 3	3.9518	3.7839	3.7839	4.2876	0.2908	[3.7839, 4.2876]	0.2149	0.2149
rho-signed_frobenius_strong	W782cd1b187b6	icv1-f2m17-tm101-00378d4e	3 / 3	4.0781	4.0377	3.9089	4.2876	0.1926	[3.9089, 4.2876]	0.2149	0.2149
rho-signed_frobenius_strong	Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	3 / 3	2.2004	2.2212	2.0963	2.2836	0.0954	[2.0963, 2.2836]	0.2033	0.2033
rho-signed_frobenius_strong	W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	3 / 3	2.6174	2.7208	2.4105	2.7208	0.1792	[2.4105, 2.7208]	0.2033	0.2033
rho-signed_frobenius_strong	W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	3 / 3	0.8769	0.8455	0.7522	1.0329	0.1429	[0.7522, 1.0329]	0.1848	0.1848
rho-signed_frobenius_strong	W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	3 / 3	0.8663	0.8767	0.8142	0.9079	0.0477	[0.8142, 0.9079]	0.1848	0.1848
rho-signed_frobenius_strong	W4aedbc100172	icv1-f2m29-tm40309-30c52b96	3 / 3	4.4930	4.3878	4.3878	4.7033	0.1821	[4.3878, 4.7033]	0.1646	0.1646
rho-signed_frobenius_strong	W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	3 / 3	4.5949	4.5431	4.3878	4.8538	0.2372	[4.3878, 4.8538]	0.1646	0.1646
rho-signed_frobenius_strong	W14592200e40d	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.1815	0.1577	0.1548	0.2319	0.0437	[0.1548, 0.2319]	0.1351	0.1351
rho-signed_frobenius_strong	W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.2363	0.2939	0.1088	0.3061	0.1106	[0.1088, 0.3061]	0.1351	0.1351
rho-signed_frobenius_strong	W46a98355e416	icv1-f2m47-t22705043-f4e44623	3 / 3	0.1235	0.1564	0.0418	0.1722	0.0712	[0.0418, 0.1722]	0.1293	0.1293
rho-signed_frobenius_strong	W606d09fccb11	icv1-f2m47-t22705043-f4e44623	3 / 3	0.0904	0.0516	0.0152	0.2044	0.1004	[0.0152, 0.2044]	0.1293	0.1293
rho-negation	W10f4a21252f1	icv1-f2m17-tm101-00378d4e	3 / 3	1.7064	1.4487	1.1519	2.5186	0.7189	[1.1519, 2.5186]	0.2149	0.8862
rho-negation	W782cd1b187b6	icv1-f2m17-tm101-00378d4e	3 / 3	1.8092	1.8587	1.3862	2.1827	0.4006	[1.3862, 2.1827]	0.2149	0.8862
rho-negation	Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	3 / 3	1.9783	1.9731	1.2354	2.7264	0.7455	[1.2354, 2.7264]	0.2033	0.8862
rho-negation	W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	3 / 3	1.4644	1.5106	1.2959	1.5867	0.1508	[1.2959, 1.5867]	0.2033	0.8862
rho-negation	W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	3 / 3	1.2742	1.4410	0.7610	1.6206	0.4534	[0.7610, 1.6206]	0.1848	0.8862
rho-negation	W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	3 / 3	1.7096	1.6470	1.5615	1.9203	0.1874	[1.5615, 1.9203]	0.1848	0.8862
rho-negation	W4aedbc100172	icv1-f2m29-tm40309-30c52b96	3 / 3	1.6954	1.6501	1.3929	2.0432	0.3275	[1.3929, 2.0432]	0.1646	0.8862
rho-negation	W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	3 / 3	2.2454	2.3247	1.2667	3.1449	0.9416	[1.2667, 3.1449]	0.1646	0.8862
rho-negation	W14592200e40d	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.4152	0.4131	0.1739	0.6587	0.2424	[0.1739, 0.6587]	0.1351	0.8862
rho-negation	W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.4512	0.2077	0.1144	1.0317	0.5049	[0.1144, 1.0317]	0.1351	0.8862
rho-negation	W46a98355e416	icv1-f2m47-t22705043-f4e44623	3 / 3	1.0520	1.2158	0.6503	1.2901	0.3499	[0.6503, 1.2901]	0.1293	0.8862
rho-negation	W606d09fccb11	icv1-f2m47-t22705043-f4e44623	3 / 3	1.1003	1.2272	0.7826	1.2912	0.2770	[0.7826, 1.2912]	0.1293	0.8862
bsgs-negation	W10f4a21252f1	icv1-f2m17-tm101-00378d4e	3 / 3	0.7146	0.7146	0.7146	0.7146	0.0000	[0.7146, 0.7146]	0.2149	1.0000
bsgs-negation	W782cd1b187b6	icv1-f2m17-tm101-00378d4e	3 / 3	1.2144	1.2144	1.2144	1.2144	0.0000	[1.2144, 1.2144]	0.2149	1.0000
bsgs-negation	Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	3 / 3	0.6538	0.6538	0.6538	0.6538	0.0000	[0.6538, 0.6538]	0.2033	1.0000
bsgs-negation	W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	3 / 3	1.0324	1.0324	1.0324	1.0324	0.0000	[1.0324, 1.0324]	0.2033	1.0000
bsgs-negation	W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	3 / 3	1.1115	1.1115	1.1115	1.1115	0.0000	[1.1115, 1.1115]	0.1848	1.0000
bsgs-negation	W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	3 / 3	0.5814	0.5814	0.5814	0.5814	0.0000	[0.5814, 0.5814]	0.1848	1.0000
bsgs-negation	W4aedbc100172	icv1-f2m29-tm40309-30c52b96	3 / 3	1.0871	1.0871	1.0871	1.0871	0.0000	[1.0871, 1.0871]	0.1646	1.0000
bsgs-negation	W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	3 / 3	1.2618	1.2618	1.2618	1.2618	0.0000	[1.2618, 1.2618]	0.1646	1.0000

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
bsgs-negation	W14592200e40d	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.5051	0.5051	0.5051	0.5051	0.0000	[0.5051, 0.5051]	0.1351	1.0000
bsgs-negation	W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	3 / 3	1.2776	1.2776	1.2776	1.2776	0.0000	[1.2776, 1.2776]	0.1351	1.0000
bsgs-negation	W46a98355e416	icv1-f2m47-t22705043-f4e44623	3 / 3	1.2618	1.2618	1.2618	1.2618	0.0000	[1.2618, 1.2618]	0.1293	1.0000
bsgs-negation	W606d09fccb11	icv1-f2m47-t22705043-f4e44623	3 / 3	1.4361	1.4361	1.4361	1.4361	0.0000	[1.4361, 1.4361]	0.1293	1.0000
kangaroo-vow	W10f4a21252f1	icv1-f2m17-tm101-00378d4e	3 / 3	1.8899	1.8899	1.8899	1.8899	0.0000	[1.8899, 1.8899]	0.2149	2.0000
kangaroo-vow	W782cd1b187b6	icv1-f2m17-tm101-00378d4e	3 / 3	1.3237	1.3237	1.3237	1.3237	0.0000	[1.3237, 1.3237]	0.2149	2.0000
kangaroo-vow	Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	3 / 3	2.4259	2.4259	2.4259	2.4259	0.0000	[2.4259, 2.4259]	0.2033	2.0000
kangaroo-vow	W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	3 / 3	0.6597	0.6597	0.6597	0.6597	0.0000	[0.6597, 0.6597]	0.2033	2.0000
kangaroo-vow	W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	3 / 3	0.7537	0.7537	0.7537	0.7537	0.0000	[0.7537, 0.7537]	0.1848	2.0000
kangaroo-vow	W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	3 / 3	2.8726	2.8726	2.8726	2.8726	0.0000	[2.8726, 2.8726]	0.1848	2.0000
kangaroo-vow	W4aedbc100172	icv1-f2m29-tm40309-30c52b96	3 / 3	0.4853	0.4853	0.4853	0.4853	0.0000	[0.4853, 0.4853]	0.1646	2.0000
kangaroo-vow	W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	3 / 3	1.0871	1.0871	1.0871	1.0871	0.0000	[1.0871, 1.0871]	0.1646	2.0000
kangaroo-vow	W14592200e40d	icv1-f2m43-tm998717-e2e742b0	3 / 3	3.2695	3.2695	3.2695	3.2695	0.0000	[3.2695, 3.2695]	0.1351	2.0000
kangaroo-vow	W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	3 / 3	2.6742	2.6742	2.6742	2.6742	0.0000	[2.6742, 2.6742]	0.1351	2.0000
kangaroo-vow	W46a98355e416	icv1-f2m47-t22705043-f4e44623	3 / 3	2.2088	2.2088	2.2088	2.2088	0.0000	[2.2088, 2.2088]	0.1293	2.0000
kangaroo-vow	W606d09fccb11	icv1-f2m47-t22705043-f4e44623	3 / 3	2.8892	2.8892	2.8892	2.8892	0.0000	[2.8892, 2.8892]	0.1293	2.0000
rho-negation-aa	W10f4a21252f1	icv1-f2m17-tm101-00378d4e	3 / 3	1.7064	1.4487	1.1519	2.5186	0.7189	[1.1519, 2.5186]	0.2149	0.8862
rho-negation-aa	W782cd1b187b6	icv1-f2m17-tm101-00378d4e	3 / 3	1.8092	1.8587	1.3862	2.1827	0.4006	[1.3862, 2.1827]	0.2149	0.8862
rho-negation-aa	Wcf0c6a05aa9c	icv1-f2m19-tm797-9c54981b	3 / 3	1.9783	1.9731	1.2354	2.7264	0.7455	[1.2354, 2.7264]	0.2033	0.8862
rho-negation-aa	W55afdcb1f85b	icv1-f2m19-tm797-9c54981b	3 / 3	1.4644	1.5106	1.2959	1.5867	0.1508	[1.2959, 1.5867]	0.2033	0.8862
rho-negation-aa	W3609e428e486	icv1-f2m23-tm5197-1f85e9e1	3 / 3	1.2742	1.4410	0.7610	1.6206	0.4534	[0.7610, 1.6206]	0.1848	0.8862
rho-negation-aa	W9b1bd6dfa11d	icv1-f2m23-tm5197-1f85e9e1	3 / 3	1.7096	1.6470	1.5615	1.9203	0.1874	[1.5615, 1.9203]	0.1848	0.8862
rho-negation-aa	W4aedbc100172	icv1-f2m29-tm40309-30c52b96	3 / 3	1.6954	1.6501	1.3929	2.0432	0.3275	[1.3929, 2.0432]	0.1646	0.8862
rho-negation-aa	W63b0bec07aa3	icv1-f2m29-tm40309-30c52b96	3 / 3	2.2454	2.3247	1.2667	3.1449	0.9416	[1.2667, 3.1449]	0.1646	0.8862
rho-negation-aa	W14592200e40d	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.4152	0.4131	0.1739	0.6587	0.2424	[0.1739, 0.6587]	0.1351	0.8862
rho-negation-aa	W99157a3b2f40	icv1-f2m43-tm998717-e2e742b0	3 / 3	0.4512	0.2077	0.1144	1.0317	0.5049	[0.1144, 1.0317]	0.1351	0.8862
rho-negation-aa	W46a98355e416	icv1-f2m47-t22705043-f4e44623	3 / 3	1.0520	1.2158	0.6503	1.2901	0.3499	[0.6503, 1.2901]	0.1293	0.8862
rho-negation-aa	W606d09fccb11	icv1-f2m47-t22705043-f4e44623	3 / 3	1.1003	1.2272	0.7826	1.2912	0.2770	[0.7826, 1.2912]	0.1293	0.8862

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-signed_frobenius_strong	W10f4a21252f1	3 / 3	1012.1	18.3855	–	–	–	–	–	5.2839	32.4490	informative, L1
rho-signed_frobenius_strong	W782cd1b187b6	3 / 3	1044.4	18.9729	–	–	–	–	–	4.4360	31.1460	informative, L1
rho-signed_frobenius_strong	Wcf0c6a05aa9c	3 / 3	1127.4	10.8225	–	–	–	–	–	5.3448	35.8740	informative, L1
rho-signed_frobenius_strong	W55afdcb1f85b	3 / 3	1341.1	12.8735	–	–	–	–	–	5.3729	33.2170	informative, L1
rho-signed_frobenius_strong	W3609e428e486	3 / 3	1796.4	4.7452	–	–	–	–	–	5.0608	32.5740	informative, L1
rho-signed_frobenius_strong	W9b1bd6dfa11d	3 / 3	1774.7	4.6879	–	–	–	–	–	5.7103	39.3910	informative, L1
rho-signed_frobenius_strong	W4aedbc100172	3 / 3	925.8	27.3017	–	–	–	–	–	5.8345	35.9460	informative, L1
rho-signed_frobenius_strong	W63b0bec07aa3	3 / 3	946.8	27.9210	–	–	–	–	–	5.0265	34.7920	informative, L1
rho-signed_frobenius_strong	W14592200e40d	3 / 3	12366.6	1.3427	–	–	–	–	–	6.5034	37.6610	informative, L1
rho-signed_frobenius_strong	W99157a3b2f40	3 / 3	16101.9	1.7483	–	–	–	–	–	6.1798	36.7580	informative, L1
rho-signed_frobenius_strong	W46a98355e416	3 / 3	40356.7	0.9554	–	–	–	–	–	9.8954	48.2120	informative, L1
rho-signed_frobenius_strong	W606d09fccb11	3 / 3	29546.0	0.6994	–	–	–	–	–	7.0914	39.0520	informative, L1
rho-negation	W10f4a21252f1	3 / 3	437.0	7.9387	15.0	245.0	–	–	–	5.4039	33.8350	informative, L1
rho-negation	W782cd1b187b6	3 / 3	463.3	8.4171	19.3	273.0	–	–	–	4.4165	29.9120	informative, L1
rho-negation	Wcf0c6a05aa9c	3 / 3	1013.7	9.7303	37.3	766.7	–	–	–	5.7221	38.2230	informative, L1
rho-negation	W55afdcb1f85b	3 / 3	750.3	7.2025	33.7	520.0	–	–	–	5.3796	33.5680	informative, L1

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-negation	W3609e428e486	3 / 3	2610.3	6.8953	62.0	2114.7	–	–	–	6.9404	44.1310	informative, L1
rho-negation	W9b1bd6dfa11d	3 / 3	3502.3	9.2515	93.7	2933.0	–	–	–	5.3889	30.7370	informative, L1
rho-negation	W4aedbc100172	3 / 3	349.3	10.3019	10.7	170.3	–	–	–	5.3053	43.7650	informative, L1
rho-negation	W63b0bec07aa3	3 / 3	462.7	13.6442	16.0	271.7	–	–	–	5.3475	36.6630	informative, L1
rho-negation	W14592200e40d	3 / 3	28297.7	3.0725	107.7	26818.3	–	–	–	12.6257	47.8330	informative, L1
rho-negation	W99157a3b2f40	3 / 3	30751.7	3.3389	115.0	29231.3	–	–	–	10.8835	38.4280	informative, L1
rho-negation	W46a98355e416	3 / 3	343782.0	8.1384	640.3	331298.0	–	–	–	109.7608	136.6990	informative, L1
rho-negation	W606d09fccb11	3 / 3	359559.0	8.5119	677.3	346648.7	–	–	–	92.3697	123.4470	informative, L1
bsgs-negation	W10f4a21252f1	3 / 3	183.0	3.3245	–	174.0	–	–	–	5.2513	43.5340	informative, L1
bsgs-negation	W782cd1b187b6	3 / 3	311.0	5.6498	–	302.0	–	–	–	5.5355	35.7800	informative, L1
bsgs-negation	Wcf0c6a05aa9c	3 / 3	335.0	3.2157	–	325.0	–	–	–	5.2680	36.2330	informative, L1
bsgs-negation	W55afdcb1f85b	3 / 3	529.0	5.0779	–	519.0	–	–	–	5.2063	36.2740	informative, L1
bsgs-negation	W3609e428e486	3 / 3	2277.0	6.0147	–	2265.0	–	–	–	5.2458	42.1900	informative, L1
bsgs-negation	W9b1bd6dfa11d	3 / 3	1191.0	3.1461	–	1179.0	–	–	–	4.5230	30.4460	informative, L1
bsgs-negation	W4aedbc100172	3 / 3	224.0	6.6058	–	215.0	–	–	–	5.7025	43.9430	informative, L1
bsgs-negation	W63b0bec07aa3	3 / 3	260.0	7.6675	–	251.0	–	–	–	6.2192	46.2700	informative, L1
bsgs-negation	W14592200e40d	3 / 3	34420.0	3.7372	–	34398.0	–	–	–	17.4184	55.1110	informative, L1
bsgs-negation	W99157a3b2f40	3 / 3	87065.0	9.4532	–	87043.0	–	–	–	28.5900	59.5770	informative, L1
bsgs-negation	W46a98355e416	3 / 3	412332.0	9.7612	–	412303.0	–	–	–	147.2592	179.6960	informative, L1
bsgs-negation	W606d09fccb11	3 / 3	469269.0	11.1091	–	469240.0	–	–	–	170.1771	200.7860	informative, L1
kangaroo-vow	W10f4a21252f1	3 / 3	484.0	8.7926	64.0	456.0	–	–	–	6.2325	35.9100	informative, L1
kangaroo-vow	W782cd1b187b6	3 / 3	339.0	6.1584	51.0	311.0	–	–	–	5.2928	33.4240	informative, L1
kangaroo-vow	Wcf0c6a05aa9c	3 / 3	1243.0	11.9317	164.0	1210.0	–	–	–	5.4144	34.5440	informative, L1
kangaroo-vow	W55afdcb1f85b	3 / 3	338.0	3.2445	50.0	305.0	–	–	–	5.4852	35.5630	informative, L1
kangaroo-vow	W3609e428e486	3 / 3	1544.0	4.0785	103.0	1505.0	–	–	–	5.1540	33.9500	informative, L1
kangaroo-vow	W9b1bd6dfa11d	3 / 3	5885.0	15.5454	359.0	5846.0	–	–	–	5.9440	34.3160	informative, L1
kangaroo-vow	W4aedbc100172	3 / 3	100.0	2.9490	7.0	69.0	–	–	–	0.0595	38.3990	informative, L1
kangaroo-vow	W63b0bec07aa3	3 / 3	224.0	6.6058	23.0	193.0	–	–	–	0.0744	34.8510	informative, L1
kangaroo-vow	W14592200e40d	3 / 3	222812.0	24.1921	1747.0	222749.0	–	–	–	50.6307	81.3590	informative, L1
kangaroo-vow	W99157a3b2f40	3 / 3	182242.0	19.7871	1409.0	182179.0	–	–	–	41.0757	72.2240	informative, L1
kangaroo-vow	W46a98355e416	3 / 3	721774.0	17.0867	2765.0	721705.0	–	–	–	167.7731	209.2150	informative, L1
kangaroo-vow	W606d09fccb11	3 / 3	944128.0	22.3505	3633.0	944059.0	–	–	–	214.5736	253.0440	informative, L1
rho-negation-aa	W10f4a21252f1	3 / 3	437.0	7.9387	15.0	245.0	–	–	–	4.6735	33.6190	informative, L1
rho-negation-aa	W782cd1b187b6	3 / 3	463.3	8.4171	19.3	273.0	–	–	–	4.9409	33.5540	informative, L1
rho-negation-aa	Wcf0c6a05aa9c	3 / 3	1013.7	9.7303	37.3	766.7	–	–	–	5.8183	35.8410	informative, L1
rho-negation-aa	W55afdcb1f85b	3 / 3	750.3	7.2025	33.7	520.0	–	–	–	5.7695	35.5050	informative, L1
rho-negation-aa	W3609e428e486	3 / 3	2610.3	6.8953	62.0	2114.7	–	–	–	6.5239	34.5510	informative, L1
rho-negation-aa	W9b1bd6dfa11d	3 / 3	3502.3	9.2515	93.7	2933.0	–	–	–	6.3731	35.7490	informative, L1
rho-negation-aa	W4aedbc100172	3 / 3	349.3	10.3019	10.7	170.3	–	–	–	5.1224	36.6290	informative, L1
rho-negation-aa	W63b0bec07aa3	3 / 3	462.7	13.6442	16.0	271.7	–	–	–	5.3396	33.6990	informative, L1
rho-negation-aa	W14592200e40d	3 / 3	28297.7	3.0725	107.7	26818.3	–	–	–	13.5097	46.2730	informative, L1
rho-negation-aa	W99157a3b2f40	3 / 3	30751.7	3.3389	115.0	29231.3	–	–	–	9.0445	41.4900	informative, L1
rho-negation-aa	W46a98355e416	3 / 3	343782.0	8.1384	640.3	331298.0	–	–	–	95.3245	131.3760	informative, L1
rho-negation-aa	W606d09fccb11	3 / 3	359559.0	8.5119	677.3	346648.7	–	–	–	94.3132	124.4010	informative, L1

Scaling

arm	sizes	verified runs	α [d5]	se(α)	log2 C	R^2
rho-signed_frobenius_strong	6	36	0.248	0.015	5.858	0.9856
rho-negation	6	36	0.430	0.029	1.967	0.9817
bsgs-negation	6	36	0.511	0.015	−0.266	0.9968
kangaroo-vow	6	36	0.562	0.021	−0.660	0.9944
rho-negation-aa	6	36	0.430	0.029	1.967	0.9817

Raw records

`research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/koblitz/records.jsonl` (240 records; ids in `raw_records.csv` and `report.json`); child stderr in `research\/ecbench\_comprehensive\_sweep\_20261018\/sessions\/koblitz/exec/` (0 file(s)).

Derivations

- **[d1]** S = cost.s of each verified, non-warm-up record (`total_gae / √r`, stored by the runner); mean, median, min, max and sample stdev (stats.rs) are over those records.
- **[d2]** 95% percentile bootstrap interval of the mean S: 4000 resamples of the verified runs of the cell with replacement (stats::bootstrap_ci, seed 20261002). A cell with one run has no interval. Not a significance test.
- **[d3]** total gae and S / floor are means over the verified runs of the cell of cost.total_gae and boundaries.ratio_to_floor; floor S is boundaries.floor_s = √(π / 2A), A = automorphisms the curve offers a generic method.
- **[d4]** expected S is methods::expected_s(method, A): the closed form the method's mean S approaches as r grows (set-up omitted). A reference mark, not a measurement.
- **[d5]** scaling: per arm, per curve, the mean gae over the arm's verified runs on that curve; the line is ordinary least squares of log2(mean gae) on log2(r) over the curves (bounds::ols), slope α, intercept log2 C so gae = C·r^α. The standard error of α is √(SS_res/(sizes−2)/S_xx) from those residuals (none below 3 sizes). Counted operations, one point per size: it is not the cluster-bootstrap interval of `ecbench bound fit`, which resamples runs and sizes.
- **[d6]** counters (distinguished points, steps) and field operations (muls, sqrs, invs) are means over the cell's verified runs of the counters the record carries (steps = `steps`, or baby + giant steps for BSGS); a dash means the method did not count it (unknown, not zero).
- **[d7]** wall and CPU are medians over the cell's verified runs of time.solve_wall_ns and time.user_ns + sys_ns. The figure is admitted only when every run behind it earned the level the spec requires (compare.rs's gate); otherwise it is informative at the lowest level earned, and never a result.
- **[d8]** phase share = the phase's mean gae over the sum of the arm's mean gae across phases on that curve, over verified runs.